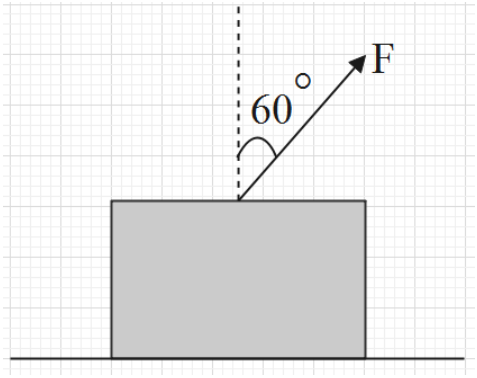
Fizika fanidan test. (14.08.2022.)

1. **20 g/sm^3** necha **kg/ dm^3** ga teng
   A) 20 B) 2000 C) 0,02 D)200

2. Massasi **10 kg** qirrasining uzunligi **25 sm** bo`lgan jismga rasmda ko`rsatilgandek **100 N** kuch ta`sir qilsa jismning asosga beradigan bosimi qanday bo`ladi? **(Pa)**



   A) 400 B) 800 C) 8000 D) 4000

3. Jismning impulsi **40 kg m/s**. Agar jism boshqa jism bilan noelastik urilishi natijasida energiyasi **51 foiz** kamaysa impulsi qanday bo`lib qoladi?
   A) 14 B) 28 C) 32 D) 16

4. **1 sm^2** yuzali jo`mrakdan **1 m/s** tezlik bilan otilib chiqayotgan suvning quvvati topilsin. **(mW)**
   A) 25 B) 50 C) 100 D) 200

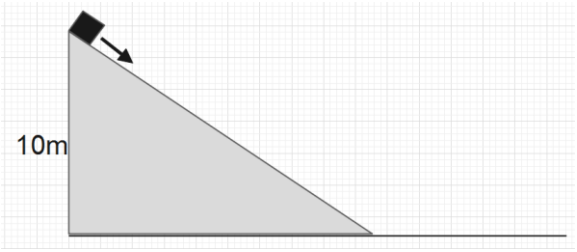
5. Diametri **2 mm** bo`lgan tomizg`ichdan tomadigan spirt tomchisi massasi qanday?**(mg)** $\pi = 3$
   A) 12 B) 24 C) 36 D) 20

6. Qattiq jismlarda tovushning tarqalish formulasini ko`rsating.
   A) $v = \sqrt{\frac{E}{\rho}}$ **B)** $v = \sqrt{\frac{\rho}{E}}$
   C) $v = \sqrt{\frac{\rho T}{E}}$ **D)** $v = \sqrt{\frac{\rho}{ET}}$

7. Prujinaga yuk osildi so`ngra prujinani yarmi kesib tashlanib yana xuddi shu yuk osildi.
   Bunda nisbiy uzayish qanday o`zgaradi?
   A) 2 marta kamayadi B) 2 marta oshadi
   C) o`zgarmaydi D) 4 marta kamayadi

8. Jism **5 sm** amplituda va **5 Hz** chastota bilan tebranmoqda. **1 s** da u necha metr masofa bosib o`tadi?
   A) 1 B) 2 C) 0,5 D) 4

9. Rasmda ko`rsatilgan jismga **10 m/s** tezlik tezlik berildi natijada jism gorizontal tekislikda qandaydir nuqtada to`xtadi. Bunda ishqalanish kuchi ishining moduli topilsin.**(m=3 kg)**



   A) 100J B) 200J C) 300J D) 450J

10. Gorizontal tekislikda yotgan **60 kg** massali yukka dastlab gorizontal **140 N** so`ngra **300 N** kuch ta`sir qildi. Ikkala holda ishqalanish kuchlari farqini toping.**(N)** $\mu = 0,5$
    A) 40 B) 160 C) 60 D) 100

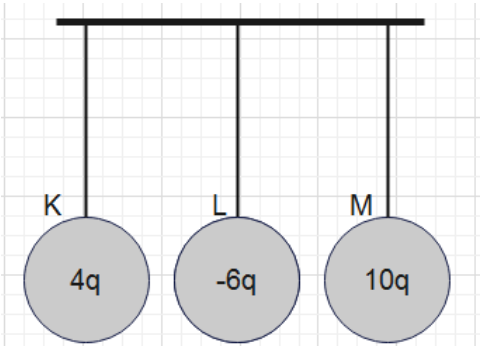
11. Ikkita zarra vakumda F kuch bilan ta`sirlashyapdi. Agar ular $\varepsilon = 4$ ga teng bo`lgan muhitga olib tushilsa ta`sir kuchi qanday bo`ladi?
    A) F/4 B) 4F C) o`zgarmaydi D) 2F

12. Idishga qamalgan geliy gazining bosimi **100 kPa** ga ortsa ichki energiya zichligi qanday o`zgaradi?**(kJ/m^3)**
    A) 300 B) 150 C) 450 D) 250

13. **5 mol** muzni eritish uchun qancha issiqlik miqdori kerak **(kJ)?**
    A) 3,7 B) 21,5 C) 42,8 D) 29,7

14. Vakumda **6q** va **-2q** miqdorda zaryadlangan ikkita bir xil sharcha ma`lum masofada joylashgan. Sharchalar bir biriga tekkizildi va $\varepsilon = 2$ ga teng bo`lgan muhitga tushirilib xuddi avvalgi masofaga joylashtirildi. Bunda ularning o`zaro ta`sir kuchi qanday o`zgaradi.
    A) 6 marta oshadi B) 6 marta kamayadi
    C) 4 marta kamayadi D) 4 marta oshadi

15. 3 ta bir xil sharcha dielektrik iplarga osib qo`yilgan. Dastlab L jism K jismga so`ngra esa L jism M jismga tekkiib olindi. Shundan so`ng M jismning zaryadi qanday o`zgargan?



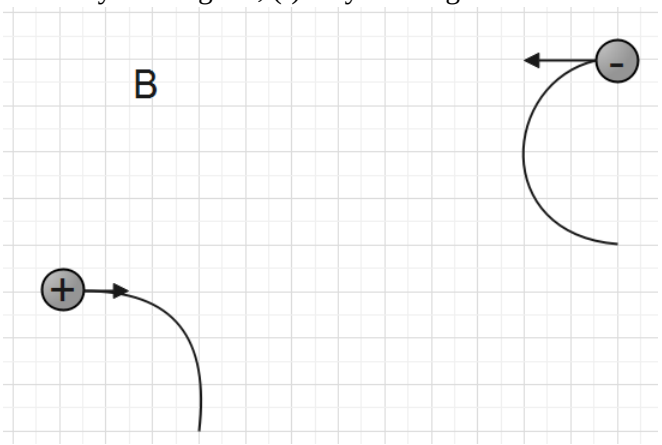
    A) 5q ga oshgan B) 5,5q ga kamaygan
    C) 5,5 q g oshgan D) 5q ga kamaygan

16. Tok kuchi ifodasini toping.
    A) $I = UR$ **B)** $I = P/R$
    C) $I = P/U$ **D)** $I = R/P$

17. O`tkazgichdan **2 A** tok o`tganda unda ajraladigan issiqliqlik miqdori **2 s** da **36 J** bo`lgan bo`lsa, o`tkazgich qarshiligini toping.**(Om)**
A) 9 B) 4,5 C) 5 D) 3

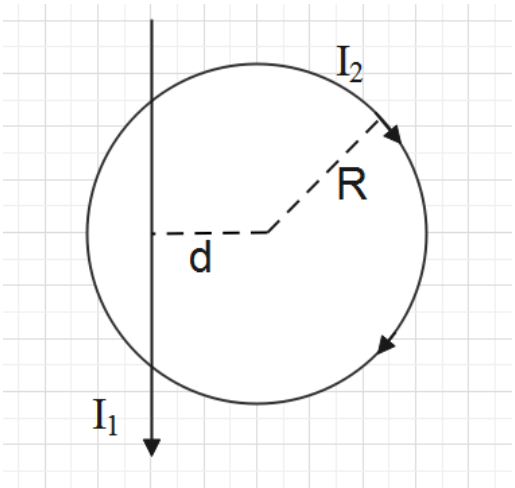
18. Vakumli diod to`yinish tokiga erishmagan. Agar anod kuchlanishi 16 marta oshirilsa tok kuchi qanday o`zgaradi?
A) 16 marta kamayadi
B) 16 marta oshadi
C) 64 marta oshadi
D) 64 marta kmayadi

19. Magnit maydonga uchib kirgan zaryadli zarralarning shu maydondagi harakat vaqtlari nisbatini toping. (+) zaryad tezligi 2v, (-) zaryad tezligi 3v.



A) 9/8 B) 18/5 C) 5/3 D) 10/3

20. Aylana markazidagi magnit maydon induksiyasini toping.(μ)
d= 2 sm, R=4 sm, I_1=2 A, I_2=5 A, $\pi = 3$



A) 75 B) 50 C) 40 D) 65

21. Tok kuchi sinus qonuniyati bo`yicha tebransa magnit maydon energiyasi qanday o`zgaradi?
A) Chiziqli B) Garmonik
C) O`zgarmaydi D) Parabolik

22. Tebranish konturidagi elektr maydon energiyasi **15 mJ** bo`lgan paytda g`altakdagi magnit maydon energiyasi **85 mJ** bo`lgan bo`lsa, konturdagi to`la energiyani toping.**(mJ)**
A) 100 B) 50 C) 70 D) 187

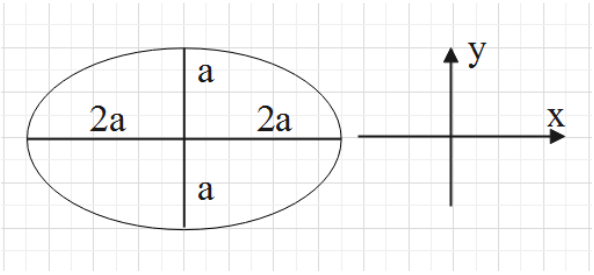
23. Obyektivining optik kuchi **2 dptr** bo`lgan fotoapparat yordamida balandligi **4 m** bo`lgan bino suratga tushirildi. Plyonkada bino tasvirining balandligi **1 sm** bo`lgan bo`lsa, bino necha metr masofadan suratga olingan?
A) 50 B) 100 C) 200 D) 40

24. Quyoshning gorizontdan balandiligi **30 gradus** edi, bir muncha vaqtdan keyin esa **60 gradus**ni tashkil etdi. Gorizontal sirtning yoritilganligi necha marta o`zgardi?
A) $\sqrt{3}$ *marta ortadi*
B)$\sqrt{3}$ *marta kamayadi*
C)$\sqrt{2}$ *marta kamayadi*
D)$\sqrt{2}$ *marta kamayadi*

25. Bemor mayda yozuvli kitobni **20 sm** dan yaxshi o`qiydi. Shifokor bemorga qanday ko`zoynak tavsiya qiladi. Sog`lom ko`zning eng yaxshi ko`rish masofasi **25 sm**.
A)2 dptr B) -2 dptr C) 1 dptr D) – 1 dptr

26. Quyoshning gorizontdan balandligi 20 gradusga teng. Yerga nisbatan necha gradusda ko`zgu qo`yilsa nur vertikal yuqoriga qaytib ketadi.
A) 35 B) 25 C) 40 D) 15

27. Ellips shaklidagi jism tinch turgan kuzatuvchiga nisbatan OX o`qi bo`yicha qanday tezlik bilan harakatlansa doira shaklida ko`rinadi?



A) $\frac{\sqrt{3}}{2}c$ B) $\frac{1}{2}c$ C) 0,6 c D) 0,8c

28. Radioaktiv atomning yarim yemirilish davri **10 minut** bo`lsa, uning **75** foizi yemirilishiga necha soat ketadi?
A) 1/2 B) 2 C) 1/3 D) 1/4

29. **1 t** massali mashina gorizontal tekislikda **a=-10 m/s^2** tezlanish tormozlanmoqda. Bunda uning yer sirtiga ko`rsatadigan to`la bosim kuchi qancha? **(kN)**
A) 10 B) 15 C) 14 D) 8

30. Polyarizatorga tushayotgan yorug`lik intensivligi $4I_0$ va polyarizator va analizator o`qlari orasidagi burchak 60^0 bo`lsa analizatordan chiqayotgan yorug`lik intensivligi qanday?
A) $\frac{I_0}{4}$ **B)** $\frac{I_0}{2}$ **C)** $\frac{I_0}{8}$ **D)** I_0